



CSA1501
· CLOUD COMPUTING AND BIG DATA ANALYTICS

The Role of Artificial Intelligence and Machine Learning in Cloud Platforms

AI/ML Cloud Services · Architecture · Industry Applications

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What Do We Mean by AI & ML in the Cloud?

Cloud platforms turn AI and ML from research disciplines into on-demand, consumable services.



Artificial Intelligence (AI)

The broad capability of a machine to perform tasks that normally require human intelligence — reasoning, perception, language, and decision-making.



Machine Learning (ML)

A subset of AI where models learn patterns directly from data and improve their performance with experience, instead of being explicitly programmed.

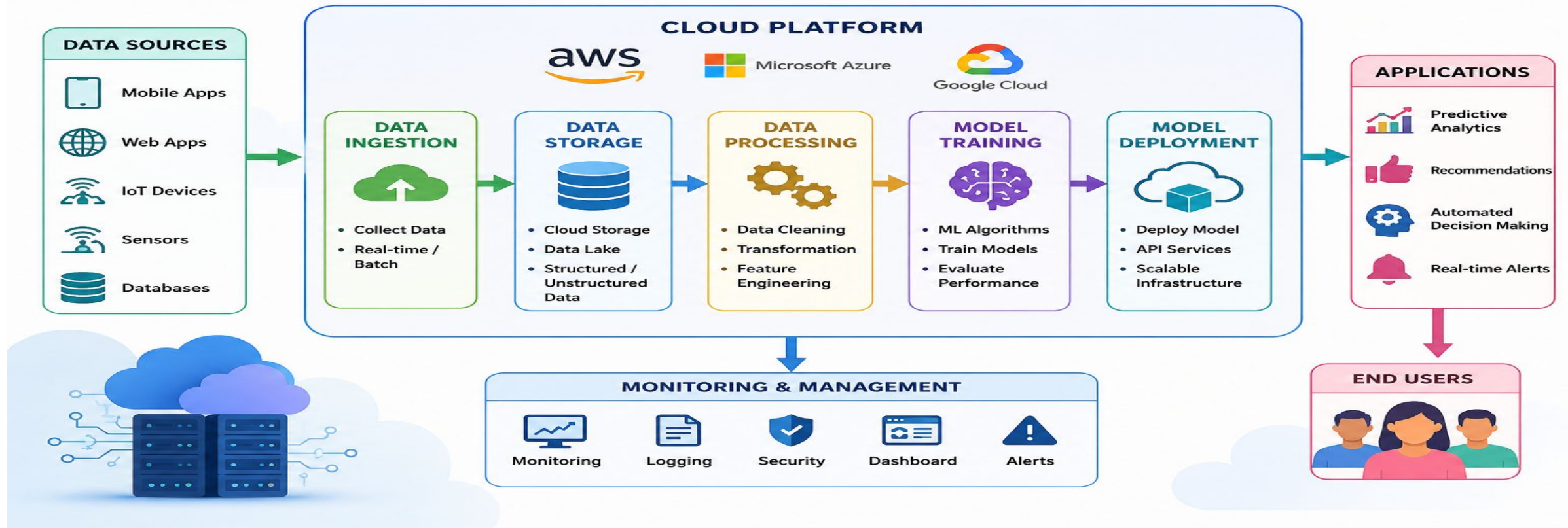


Cloud AI/ML Platform

A managed environment that supplies the compute, storage, and pre-built tools needed to build, train, and deploy AI/ML models without owning hardware.

ARCHITECTURE

AI/ML CLOUD PLATFORM ARCHITECTURE



The Cloud AI/ML Service Stack

Major providers (AWS, Google Cloud, Microsoft Azure) package AI/ML capability at three levels of abstraction.



AI Infrastructure (IaaS)

e.g. GPU/TPU instances, AWS EC2 P-series, Google Cloud TPU

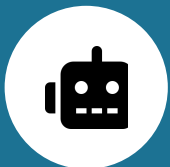
Raw, high-performance compute and storage for training large models. Users manage the software stack themselves.



ML Platforms (PaaS)

e.g. Amazon SageMaker, Google Vertex AI, Azure ML Studio

End-to-end managed environments to build, train, tune, and deploy custom models without managing servers.



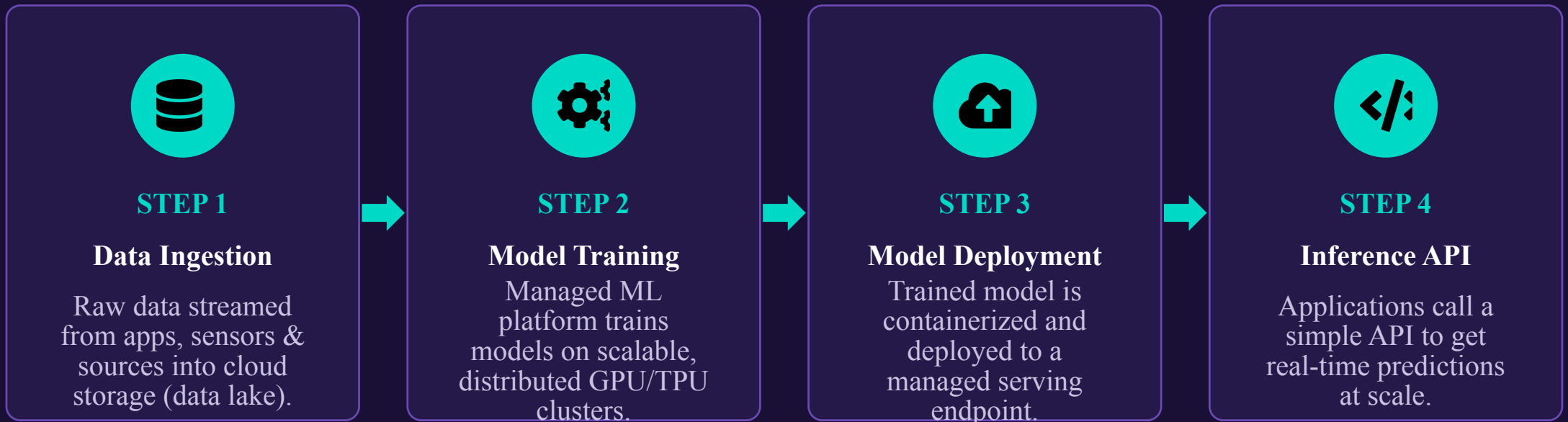
AI Services (SaaS)

e.g. Amazon Rekognition, Google Vision API, Azure OpenAI

Ready-made, pre-trained APIs for vision, language, and speech — consumed directly with no ML expertise required.

How It Works: The AI/ML Cloud Pipeline

From raw data to a live prediction, delivered as an API call.



Industry Relevance & Applications

Cloud AI/ML removes the infrastructure barrier, letting every industry adopt intelligent systems at scale.



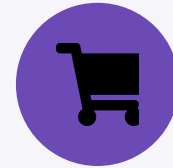
Healthcare

Cloud ML models analyze medical scans and flag anomalies for radiologists in seconds.



Finance

Real-time fraud detection APIs score millions of transactions per second for banks.



Retail

Recommendation engines personalize product suggestions using cloud-hosted models.



Autonomous Systems

Cloud-trained perception models are compressed and pushed to edge devices in vehicles.

Practical Example: Netflix on AWS Cloud AI



Personalized Recommendations at Global Scale

Challenge: Recommend the right title, out of thousands, to 260M+ subscribers in real time.

Cloud AI Stack: Runs ML training and inference on AWS compute (EC2, SageMaker-style pipelines) and cloud storage for viewing data.

How it works: Viewing history and behavior data feed ML models that continuously retrain on cloud infrastructure to update recommendations.

Result: Personalized rows and thumbnails generated per user, credited with saving over \$1B annually in reduced subscriber churn.

80%

of content watched is discovered through AI-driven recommendations

260M+

subscribers served personalized results simultaneously

\$1B+

estimated annual value from reduced churn



Conclusion & Key Takeaway

Cloud platforms have turned AI and ML from a specialized, capital-intensive discipline into an on-demand utility — accessible through simple APIs and managed services.

- ✓ Three layers — IaaS, PaaS, SaaS — let organizations choose their level of control vs. convenience.
- ✓ A common pipeline (ingest → train → deploy → infer) powers everything from fraud detection to recommendations.
- ✓ Every major industry now embeds cloud AI/ML directly into its core products.

Thank You